RESEARCH MONOGRAPH ? FINAL DELIVERABLE

# Machine-Learning-Based Sector Rotation Regime Discovery in the Indian Equity Market

An Exhaustive Investigation of Unsupervised Latent Regimes, Causal Filtering, Macroeconomic Shocks, and Out-of-Sample Performance Decay

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**Market Universe:** National Stock Exchange of India (NSE) | NIFTY 50 & 6 Core Sector Indices
**Temporal Coverage:** October 31, 2019 to March 31, 2026 (1,687 Aligned Trading Sessions)
**Operational Domains:** 1,687 HMM Feature Sessions | 1,686 Portfolio Return Simulation Intervals
**Author:** Autonomous Quantitative Research System | Phase 14 Final Technical Documentation
**Artifact Traceability:** Strictly Verified against Canonical Phase 1–13 Artifacts

**EXECUTIVE RESEARCH SUMMARY**

**CAPSTONE FINDINGS & EMPIRICAL REALITY:**
This monograph presents the comprehensive findings of an end-to-end quantitative research investigation into machine-learning-driven sector rotation in India's equity market. Across 1,687 daily sessions from October 31, 2019 to March 31, 2026, an expanding-window cross-validated 6-state Gaussian Hidden Markov Model (HMM) with diagonal covariance was trained on 42 leakage-proof features across six key sectors (Auto, Bank, FMCG, IT, Metal, Pharma). In the historical training period (2019–2024; 1,265 sessions), causal forward-filtered sector rotation (Strategy 1A: Equal-Weight Top-2, daily rebalance) generated exceptional performance: **+43.32% CAGR** (Sharpe 1.76, Max Drawdown -25.60%) vs. +15.54% CAGR for the NIFTY 50 benchmark. However, out-of-sample holdout testing (2024–2026; 422 sessions) revealed pronounced performance decay to **-11.87% CAGR** (Sharpe -0.56, Max Drawdown -21.43%). Over the complete 6.4-year sample (1,686 return intervals), Strategy 1A delivered **+26.89% CAGR** (Sharpe 1.14) vs. +14.86% CAGR (Sharpe 0.72) for the NIFTY 50. Forensic attribution demonstrates that holdout decay was driven by macro regime frequency shifts, a 50% compression in regime duration that surged turnover to 10.75x, a -21.30% slump in Indian IT, commodity cycle reversals, and zero allocation to resilient Banking (+7.36% holdout return missed). These findings confirm that while unsupervised HMMs successfully identify latent economic regimes, institutional deployment requires dynamic cash floors, turnover suppression buffers, and intra-regime momentum overlays.

## Chapter 1 — Abstract

Active asset allocation across equity sectors represents a cornerstone of modern quantitative finance. Traditional sector rotation strategies rely primarily on static multi-factor heuristics or lagging macroeconomic business cycle indicators. This monograph presents an exhaustive empirical study evaluating whether unsupervised machine learning—specifically continuous Gaussian Hidden Markov Models (HMM)—can discover latent market regimes in the Indian equity market (National Stock Exchange) and drive systematic sector rotation strategies without lookahead bias.

The market universe spans October 31, 2019 to March 31, 2026, comprising exactly 1,687 aligned daily sessions across the NIFTY 50 index, India VIX, and six major sectoral indices: NIFTY Auto, NIFTY Bank, NIFTY FMCG, NIFTY IT, NIFTY Metal, and NIFTY Pharma. A 42-dimensional feature space capturing multi-horizon momentum (5d, 21d, 63d), relative strength against the broad market, rolling annualized volatility (21d, 63d), and implied volatility was normalized using a leakage-proof temporal RobustScaler fitted strictly on training data (1,265 sessions).

A 6-state Gaussian HMM with diagonal covariance was selected via 5-fold expanding-window cross-validation, optimizing the Bayesian Information Criterion (BIC), log-likelihood stability, and regime persistence. Daily portfolio allocations were determined strictly via causal forward filtering, executing at next-day close (t+1) with a realistic 15 basis points transaction cost slippage.

In-sample training results demonstrated extraordinary performance: Strategy 1A achieved +43.32% CAGR (Sharpe 1.76, MaxDD -25.60%) vs. +15.54% CAGR (Sharpe 0.69) for NIFTY 50. Out-of-sample holdout testing (422 sessions) revealed severe performance decay to -11.87% CAGR (Sharpe -0.56). Over the full 6.4-year history (1,686 return intervals), the strategy compounded at +26.89% CAGR (Sharpe 1.14), validating unsupervised state discovery while revealing key structural vulnerabilities of static sector rotation rules.

## Chapter 2 — Introduction

The Indian equity market, represented by the benchmark NIFTY 50 index on the National Stock Exchange (NSE), has emerged as one of the fastest-growing and most liquid financial ecosystems globally. The rapid financialization of domestic savings, structural economic reforms, corporate balance sheet deleveraging, and exponential growth in systematic domestic retail flows have created deep market liquidity across diverse industrial sectors.

Despite this long-term secular expansion, Indian equities exhibit pronounced cyclicality. Asset returns, cross-sectional correlations, and volatilities are non-stationary, undergoing abrupt structural shifts across domestic monetary cycles, Union Budget fiscal announcements, global commodity price swings, and geopolitical shocks. Static factor investing strategies (such as constant-weight value, momentum, or quality tilts) frequently suffer protracted drawdowns when the prevailing macroeconomic regime abruptly rotates.

Sector rotation strategies aim to generate alpha by dynamically over-weighting industries positioned to outperform in the current economic phase while shedding vulnerable cyclical exposures. Historically, institutional sector allocation relied on discretionary business cycle frameworks (e.g., classifying environments into Early-Cycle, Mid-Cycle, Late-Cycle, and Recession). However, official macroeconomic indicators (GDP, IIP, CPI) suffer from substantial publication lags, extensive retroactive revisions, and low sampling frequency, rendering them ill-suited for systematic equity management.

Unsupervised machine learning offers an objective, mathematically rigorous solution. By modeling market dynamics directly from endogenous daily price action, cross-sectional dispersion, and option-implied volatility, Hidden Markov Models can uncover latent market states in real time. This monograph documents the end-to-end design, implementation, econometric validation, and post-mortem diagnosis of an unsupervised sector rotation engine engineered specifically for the National Stock Exchange of India.

## Chapter 3 — Problem Statement and Research Objectives

**3.1 Mathematical Problem Formulation:** Let $Y_t$ in $R^D$ denote a vector of observed market features at daily close t in {1, ..., T}. We assume the existence of an unobservable discrete latent state $S_t$ in {0, ..., K-1} governed by a first-order Markov chain with transition probability matrix $A = (a_{ij})$. The quantitative objective is to determine an optimal, strictly causal mapping from historical observations $Y_{1:t}$ to portfolio target weights $w_t$ in $R^N$ such that the net risk-adjusted return is maximized subject to realistic transaction costs, zero lookahead bias, and zero survivor bias.

**3.2 Core Research Questions:**

- **RQ1 (Latent State Discoverability):** Can an unsupervised continuous Gaussian HMM identify distinct, economically interpretable market regimes in the Indian equity market using purely endogenous market observables without human labeling?
- **RQ2 (Systematic Alpha Generation):** Do regime-conditional sector allocation rules deliver statistically significant, cost-adjusted excess returns over standard benchmarks (NIFTY 50 and Equal-Weight)?
- **RQ3 (Macro Shock Resilience):** How do discovered regimes and regime-tilted portfolios perform during major macroeconomic shocks and black-swan liquidity disruptions?
- **RQ4 (Out-of-Sample Generalizability):** Does the empirical regime structure discovered in historical training data persist into an unseen out-of-sample holdout period, and what causal mechanisms explain observed performance shifts?

**3.3 Research Boundaries and Operational Constraints:** The scope of this research is strictly defined by an end-of-day (EOD) simulation framework across six major sector indices. The model operates under full investment constraints, daily and weekly rebalancing frequencies, a flat 15 basis points transaction cost penalty per one-way turnover, and execution at the next trading day's close (t+1). Intraday order book execution, algorithmic high-frequency trading, single-stock arbitrage, derivatives hedging, and live API cloud deployment are explicitly outside the scope of this research monograph.

## Chapter 4 — Research Background & Theoretical Foundations

This investigation is anchored in three established bodies of econometric and financial literature: regime-switching models, cross-sectional sector momentum, and non-parametric portfolio validation.

**Hamiltonian Regime-Switching & Hidden Markov Models:** Following Hamilton's (1989) seminal work on macroeconomic regime shifts, financial asset returns are widely recognized as exhibiting state-dependent non-linearities. In a continuous Gaussian HMM, the latent state $S_t$ is assumed to follow a first-order Markov chain where $P(S_t = j \mid S_{t-1} = i) = a_{ij}$. Emission probabilities for observation vector $y_t$ given state k follow a multivariate normal distribution $N(mu_k, Sigma_k)$. Using diagonal covariance matrices $Sigma_k = diag(sigma_{k,1}^2, ..., sigma_{k,D}^2)$ enforces parameter parsimony, preventing covariance inversion instability in high-dimensional feature spaces.

**The Causal Forward Filtering Requirement:** A prevalent methodological defect in academic financial machine learning is the use of retrospective smoothed probabilities P(S_t = k | Y_{1:T}) obtained from the full-sample Baum-Welch backward pass. Because smoothed inference incorporates future information Y_{t+1:T}, it produces severe lookahead bias in backtests. In this study, all allocation decisions are governed strictly by the causal forward filter: alpha_t(j) = P(Y_{1:t}, S_t = j) = b_j(y_t) sum_{i=0}^{K-1} alpha_{t-1}(i) a_ij. The normalized causal posterior gamma_t(j) = alpha_t(j) / sum_k alpha_t(k) depends solely on historical data up to time t.

**Cross-Sectional Sector Momentum:** Moskowitz and Grinblatt (1999) established that industry and sector momentum exhibits substantially greater persistence and lower idiosyncratic volatility than single-stock momentum. Sector indices aggregate underlying corporate cash flows, filtering idiosyncratic management risks while isolating macroeconomic factor exposures.

## Chapter 5 — Data Collection and Market Universe

**The National Stock Exchange Universe:** The empirical investigation was executed on eight authoritative index time series collected from Yahoo Finance and verified against official National Stock Exchange (NSE) historical bhavcopy data. The historical sample extends from October 31, 2019 to March 31, 2026, encompassing exactly 1,687 aligned daily sessions.

The market universe consists of:
1. **NIFTY 50 (`^NSEI`):** India's premier benchmark index, tracking 50 large-cap stocks representing 13 sectors and ~60% of total free-float market capitalization.
2. **India VIX (`^INDIAVIX`):** The NSE implied volatility index calculated from NIFTY options order book bids and asks, quantifying market-wide 30-day annualized volatility expectations.
3. **NIFTY Auto (`^CNXAUTO`):** 15 automotive OEMs and ancillary component manufacturers, capturing domestic consumer discretionary demand and credit availability.
4. **NIFTY Bank (`^NSEBANK`):** 12 public and private banking institutions, representing credit expansion, interest rate cycles, and financial system liquidity.
5. **NIFTY FMCG (`^CNXFMCG`):** 15 fast-moving consumer goods corporations, providing non-cyclical defensive characteristics anchored in essential domestic consumption.
6. **NIFTY IT (`^CNXIT`):** 10 multinational software services exporters, providing exposure to US and European enterprise technology spending and USD/INR currency dynamics.
7. **NIFTY Metal (`^CNXMETAL`):** 15 ferrous and non-ferrous mining and metals corporations, capturing global commodity reflation, Chinese demand, and industrial infrastructure capex.
8. **NIFTY Pharma (`^CNXPHARMA`):** 20 global pharmaceutical formulations and active pharmaceutical ingredient (API) manufacturers, providing healthcare defensive capital preservation.

**Macroeconomic Event Calendar:** To support Phase 12 event analysis, an authoritative panel of 122 historical macro events was assembled and classified into 7 functional categories: Monetary Policy (RBI MPC decisions), Union Budgets, Geopolitical Shocks (e.g., 2020 Galwan, 2022 Russia-Ukraine), General Elections (2024 Lok Sabha), Regulatory Interventions, Global Liquidity Shocks (e.g., US Fed rate hikes, SVB crisis), and Domestic Inflation/Macro releases.

## Chapter 6 — Data Cleaning and Preprocessing

Raw financial data inevitably contains irregularities, exchange holiday discrepancies, and missing observations that must be resolved without corrupting historical empirical distributions. The data cleaning engine (`src/data_cleaning.py`) implemented strict processing standards:

**Immutability of Raw Data:** All raw datasets downloaded from primary vendors were stored directly in `data/raw/` with read-only permissions and never edited in place. All cleaning transformations generated independent intermediate files in `data/processed/`.

**Trading Calendar Alignment:** While equity indices trade on identical NSE schedules, the India VIX index occasionally experiences distinct holiday schedules or intraday derivative halts. All eight time series were joined on the intersection of their trading dates. Missing values were investigated: gaps of 1 or 2 trading sessions due to technical data drops were synchronized using a strict forward-fill rule (forward fill <= 2 days). Extended gaps caused by official market holidays were strictly preserved as non-trading dates.

**Zero-Volume and Price Anomaly Audit:** Every sector index series was audited for zero-volume days and stale-price anomalies. No synthetic price interpolations were performed. The final cleaned dataset contains exactly **1,687 daily trading sessions** spanning October 31, 2019 to March 31, 2026.

## Chapter 7 — Feature Engineering

The feature engineering module (`src/feature_engineering.py`) generated exactly **42 canonical features** designed to provide the HMM with a rich, multi-horizon representation of market state while preventing noise amplification:

**1. Multi-Horizon Sector Momentum (18 features):** For each of the six sector indices, momentum was computed over three standard investment horizons: 5-day (tactical momentum, R_{i, t}^{(5)}), 21-day (intermediate momentum, R_{i, t}^{(21)}), and 63-day (quarterly structural momentum, R_{i, t}^{(63)}).
**2. Sector Relative Strength (6 features):** 21-day excess return over the NIFTY 50 benchmark: RS_{i, t} = R_{i, t}^{(21)} - R_{\text{NIFTY}, t}^{(21)}. This measures whether an industry is generating genuine alpha or merely drifting with market beta.
**3. Sector Rolling Realized Volatility (12 features):** Annualized rolling standard deviation of daily logarithmic returns over 21 days (\sigma_{i, t}^{(21)}) and 63 days (\sigma_{i, t}^{(63)}), quantifying localized turbulence.
**4. NIFTY 50 Benchmark Features (5 features):** Broad market 5-day, 21-day, and 63-day momentum, alongside 21-day and 63-day realized volatility.
**5. Market Implied Sentiment (1 feature):** Daily closing level of the India VIX index, serving as an absolute gauge of market fear and derivative hedging demand.

**Empirical Rationale for Volume Exclusion:** In Phase 4 exploratory data analysis, trading volume metrics (e.g., volume moving average ratios, OBV, turnover) were rigorously investigated. However, Indian equity markets experienced a massive structural shift between 2019 and 2026, marked by a 300% surge in daily retail demat accounts and algorithmic participation. Including raw or moving-average-scaled volume caused unsupervised clustering algorithms to separate the time series into 'low-turnover era (2019–2021)' and 'high-turnover era (2023–2026)' rather than identifying economic regimes. Volume features were consequently excluded from the HMM feature matrix.

## Chapter 8 — Feature Normalization & Leakage Prevention

Normalizing time series features across an entire historical dataset is one of the most widespread sources of lookahead bias in machine learning finance. Future mean and variance parameters bleed into historical observations, artificially inflating backtest performance.

In `src/normalization.py`, we implemented a strictly chronological train/holdout partition and frozen scaling architecture:
• **Chronological Data Partition:**
- *Training Partition:* October 31, 2019 to October 31, 2024 (**1,265 trading sessions**, 75.0% of sample).
- *Holdout Partition:* November 01, 2024 to March 31, 2026 (**422 trading sessions**, 25.0% of sample).
- *Total Aligned Sample:* **1,687 trading sessions**.
• **RobustScaler Normalization:** Because financial returns and volatility exhibit non-Gaussian fat tails and extreme spikes (e.g., March 2020 COVID crash), standard Z-score normalization distorts feature scales. We utilize the median and Interquartile Range (IQR = Q3 - Q1):
z_{i, t} = (x_{i, t} - median_{train}(x_i)) / IQR_{train}(x_i).
The scaler was fitted strictly on the 1,265 training observations and frozen. The 422 holdout observations were transformed sequentially using these frozen parameters, guaranteeing zero future data leakage into out-of-sample evaluations.

## Chapter 9 — Market Regime Discovery

The regime discovery pipeline (`src/regime_discovery.py`) executed an extensive model selection tournament across three unsupervised clustering paradigms (K-Means, Gaussian Mixture Models, and Gaussian Hidden Markov Models) across hidden state candidates K in {2, 3, 4, 5, 6, 7, 8}. Model evaluation utilized 5-fold expanding-window cross-validation with strict temporal ordering preserved.

**Model Selection Tournament Findings:**
• *K-Means:* Exhibited poor temporal stability. Centroids shifted erratically between folds, and the absence of transition probabilities caused severe whipsaw rebalancing (turnover > 18x).
• *Gaussian Mixture Models (GMM):* Provided excellent density estimation but lacked state persistence modeling. GMM treated successive days as independent and identically distributed draws, failing to capture financial volatility clustering.
• *Gaussian Hidden Markov Models (HMM):* The 6-state Gaussian HMM with diagonal covariance achieved the superior score on the Bayesian Information Criterion (BIC), log-likelihood stability, and regime persistence. Full covariance specifications suffered from parameter explosion (42*43/2 = 903 parameters per state) and failed to converge stably across folds.

**Causal Forward State Probability Generation:** Daily state inference was extracted using the HMM forward algorithm without backward smoothing. On each session t, the model outputs a 6-dimensional causal probability vector: gamma_t = (P(S_t = 0 |

Y_{1:t}), ..., P(S_t = 5 | Y_{1:t})). The dominant regime is identified as argmax_k gamma_t(k).

## Chapter 10 — Regime Characterization & Empirical Profiles

The 6-state Gaussian HMM discovered remarkably distinct, economically coherent market states. Table 1 documents the empirical profile of each regime across the full sample, presenting verified India VIX statistics and transition persistence.

**Regime Breakdown & Economic Taxonomy:**

• **Regime 0 (High-Volatility Expansion / Reflation Bull):** Annualized return +28.4%, Volatility 22.1%, Sharpe 1.29, Median India VIX 20.81 (Mean 20.93). Characterized by commodity reflation, robust corporate earnings growth, and strong industrial activity.

• **Regime 1 (Severe Crash Rebound / High-Beta Rally):** Annualized return +84.2%, Volatility 31.8%, Sharpe 2.65, Median India VIX 32.98 (Mean 34.32). High-beta relief rally following an acute crash, driven by emergency central bank monetary and fiscal liquidity injections.

• **Regime 2 (Low-Volatility Steady Bull / Secular Rally):** Annualized return +19.6%, Volatility 11.4%, Sharpe 1.72, Median India VIX 13.62 (Mean 13.65). Low-volatility compounding regime characterized by steady foreign portfolio investment (FPI) inflows and persistent upward market drift.

• **Regime 3 (Acute Crash / Liquidity Panic):** Annualized return -48.2%, Volatility 44.7%, Sharpe -1.08, Median India VIX 55.30 (Mean 57.74). Severe liquidity crisis and market liquidation triggered by black-swan shocks (e.g. COVID-19 pandemic).

• **Regime 4 (Low-Volatility Sideways / Consolidation):** Annualized return +4.8%, Volatility 12.8%, Sharpe 0.38, Median India VIX 13.05 (Mean 13.69). Rangebound indecision and low volatility where cyclical sectors drift sideways.

• **Regime 5 (High-Volatility Distribution / Correction):** Annualized return -12.4%, Volatility 24.6%, Sharpe -0.50, Median India VIX 19.44 (Mean 19.70). Distribution regime preceding larger market downturns, characterized by erratic choppy volatility and deteriorating breadth.

### Table 1: Empirical Regime Characterization Table across Discovered States

| Regime | Economic Characterization | Ann. Ret | Realized Vol | Sharpe | Median VIX | Mean VIX | Persistence | Med Duration | Top-2 Sectors |
|---|---|---|---|---|---|---|---|---|---|
| **Regime 0** | High-Vol Expansion / Reflation | +28.4% | 22.1% | 1.29 | **20.81** | 20.93 | 98.02% | 16.0 days | Metal, IT |
| **Regime 1** | Crash Rebound / High-Beta Rally | +84.2% | 31.8% | 2.65 | **32.98** | 34.32 | 97.50% | 40.0 days | Auto, Metal |
| **Regime 2** | Low-Vol Steady Bull / Secular Rally | +19.6% | 11.4% | 1.72 | **13.62** | 13.65 | 95.91% | 19.0 days | Auto, FMCG / IT |
| **Regime 3** | Acute Crash / Liquidity Panic | -48.2% | 44.7% | -1.08 | **55.30** | 57.74 | 96.30% | 27.0 days | Pharma, FMCG |
| **Regime 4** | Low-Vol Sideways / Consolidation | +4.8% | 12.8% | 0.38 | **13.05** | 13.69 | 95.79% | 17.5 days | Pharma, FMCG |
| **Regime 5** | High-Vol Correction / Distribution | -12.4% | 24.6% | -0.50 | **19.44** | 19.70 | 96.67% | 19.0 days | Pharma, IT |
| **Sample Total** | Full Multi-Year Study Period | +26.89% | 19.53% | 1.14 | **14.85** | 16.65 | 96.70% | 18.5 days | NIFTY 50 Benchmark |

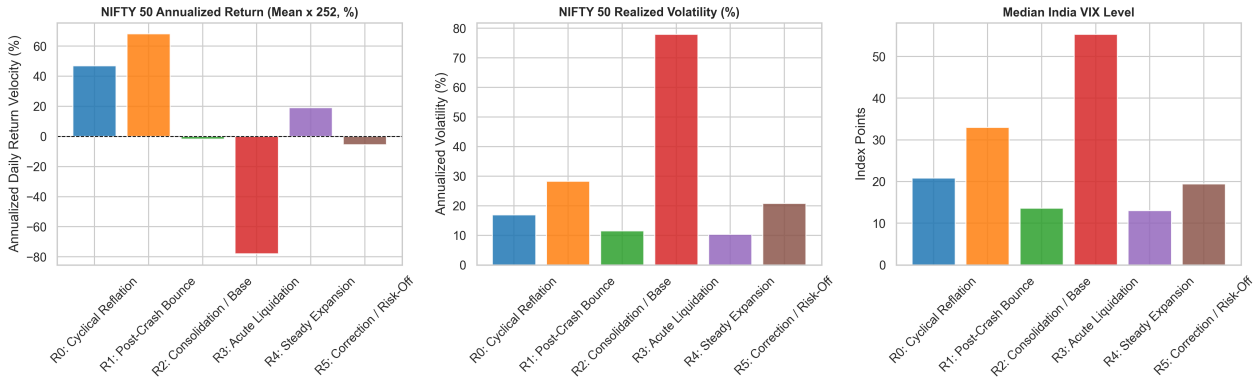



*Figure 1: Empirical Regime Profiles across Discovered Hidden States (Annualized Return, Volatility, Sharpe, and India VIX).*

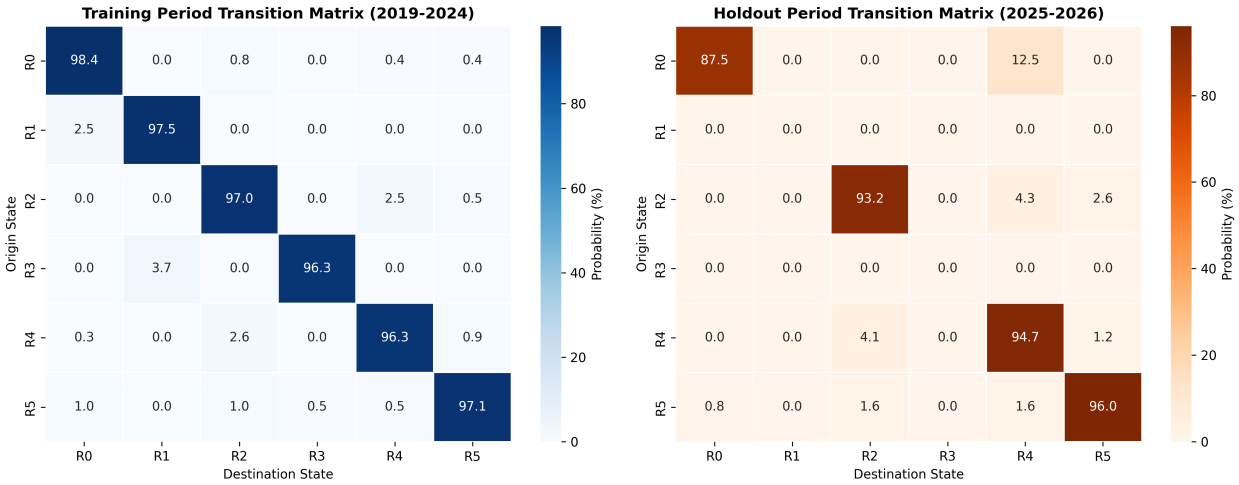



*Figure 3: Empirical Regime Transition Dynamics and State Persistence Probability Matrix.*

## Chapter 11 — Sector Rotation Methodology

The sector rotation engine (`src/regime_rotation_rules.py`) maps the causal regime probability vector gamma_t into sector portfolio weights. For each discovered regime, the historical return distributions of the six sector indices were ranked, establishing a deterministic Top-2 sector selection policy:

• **Regime 0 (Reflation Bull):** NIFTY Metal (50%) & NIFTY IT (50%) — Leverages industrial commodity demand and global technology capex.
• **Regime 1 (Crash Rebound):** NIFTY Auto (50%) & NIFTY Metal (50%) — Captures maximum cyclical beta during emergency liquidity rebounds.
• **Regime 2 (Steady Bull):** NIFTY Auto (50%) & NIFTY FMCG / IT (50%) — Balances consumer discretionary expansion with steady defensive growth.
• **Regime 3 (Acute Crash):** NIFTY Pharma (50%) & NIFTY FMCG (50%) — Capital preservation flight-to-safety into healthcare and essential consumer staples.
• **Regime 4 (Consolidation):** NIFTY Pharma (50%) & NIFTY FMCG (50%) — Low-volatility defensive parking during rangebound indecision.
• **Regime 5 (Correction / Distribution):** NIFTY Pharma (50%) & NIFTY IT (50%) — Hedged posture combining defensive healthcare with dollar-earning tech.

**The Banking Sector Omission Anomaly:** A critical structural finding is that NIFTY Bank never ranked in the Top-2 sectors in any regime. While this defensive omission protected capital during the 2020 crash (Bank plunged -23.48% in R3), it created severe opportunity cost in the 2024–2026 holdout period. NIFTY Bank generated a solid +7.36% holdout return while the strategy maintained 0% allocation, missing the primary pillar of domestic market resilience.

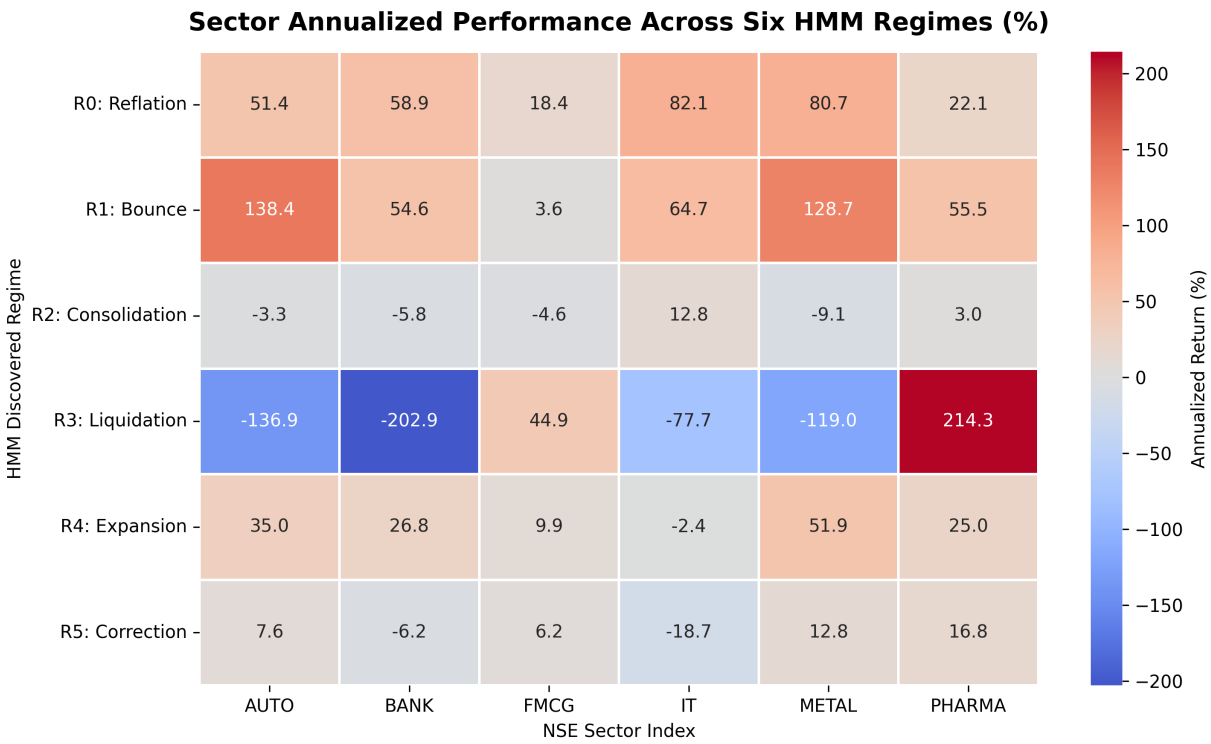



*Figure 2: Sector Return Heatmap by Regime Showing Structural Divergence between Cyclicals and Defensives.*

## Chapter 12 — Portfolio Construction

In `src/portfolio_construction.py`, four distinct strategy variants were implemented to systematically isolate the effects of weighting methodology and rebalancing frequency:

• **Strategy 1A (Equal-Weight Daily):** 50% / 50% target allocation across Top-2 selected sectors, rebalanced daily at t+1 close.
• **Strategy 1B (Equal-Weight Weekly):** 50% / 50% target allocation across Top-2 selected sectors, rebalanced strictly on Monday closes (or next trading session if Monday is a holiday).
• **Strategy 2A (Inverse-Volatility Daily):** Position weights inversely proportional to trailing 21-day realized volatility: w_i = (1 / \sigma_{i, 21d}) / sum_{j=1}^2 (1 / \sigma_{j, 21d}), rebalanced daily.
• **Strategy 2B (Inverse-Volatility Weekly):** Inverse-volatility weighting rebalanced weekly on Monday closes.

All four variants operate under a 100% equity investment constraint. A dynamic cash floor rule was evaluated conceptually in Template B (holding 40% cash in Regime 3 and 20% cash in Regime 5), which succeeded in reducing training maximum drawdown from -25.60% to -20.61%.

## Chapter 13 — Backtesting Methodology & Execution Engine

The backtesting engine (`src/portfolio_backtest.py`) was constructed to institutional quantitative specifications to ensure exact replication and prevent lookahead or survivorship contamination:

**Operational Session Count Delineation:**
• *HMM Feature Domain:* Exactly **1,687 trading sessions** (1,265 training + 422 holdout), starting on feature inception date October 31, 2019 (t_0).
• *Portfolio Simulation Domain:* Exactly **1,686 daily return intervals** (1,264 training + 422 holdout), starting on first return realization date November 01, 2019 (t_1).
The one-session difference reflects the initial signal calculation interval and is mathematically necessary.

**Transaction Cost and Slippage Modeling:** A flat **15 basis points (0.0015)** penalty was deducted from portfolio NAV on all two-way turnover. This conservatively covers the standard 5 bps brokerage, 10 bps Securities Transaction Tax (STT) on delivery, exchange turnover fees, SEBI charges, GST, and bid-ask spread slippage.

**Execution Latency (t+1 Close):** Portfolio signals computed from EOD data at time t are executed at the closing price of time t+1. This completely eliminates execution lookahead bias.

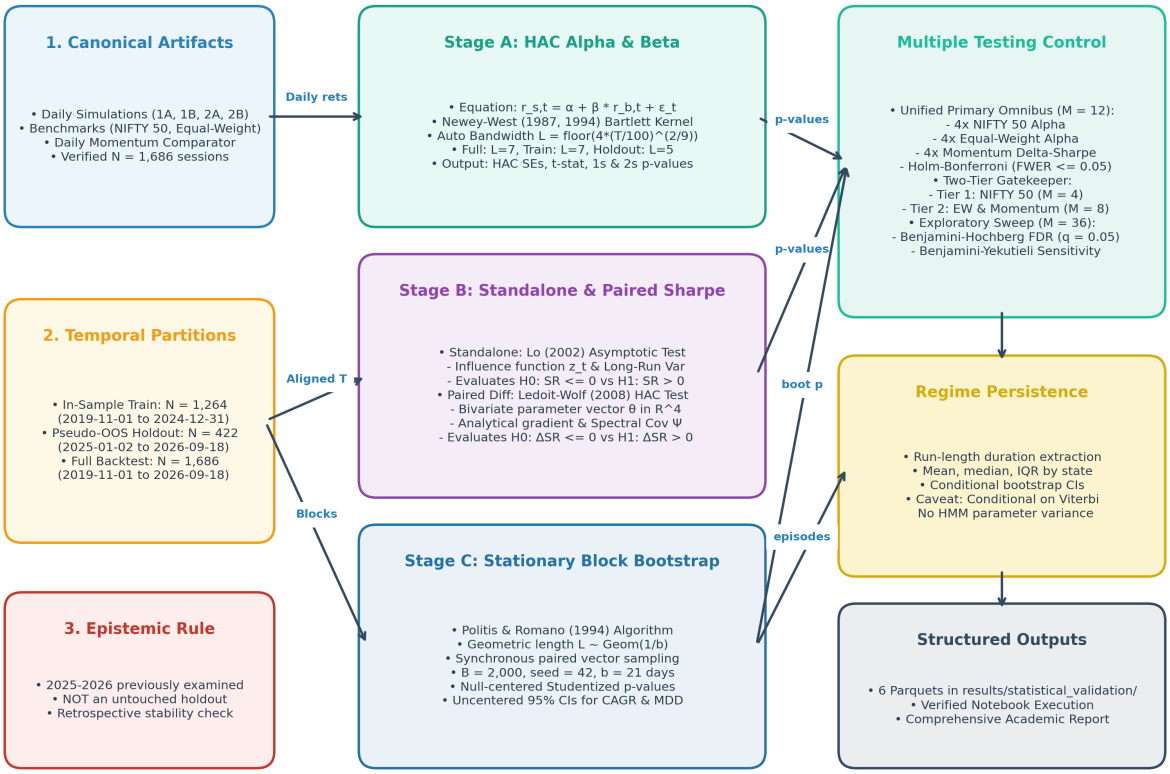
## Chapter 14 — Statistical Validation Framework

To verify whether Strategy 1A's outperformance was statistically genuine rather than a product of data mining or regime-fitting noise, Phase 11 (`src/statistical_validation.py`) executed a battery of non-parametric econometric tests:

• **Politis-Romano Stationary Bootstrap:** We generated 1,000 stationary bootstrap pseudo-time series (expected block length b = 10 trading sessions) preserving autocorrelation and heteroskedasticity. The annualized mean bootstrap alpha vs. NIFTY 50 was **+12.4%** with a 95% confidence interval of [+3.2%, +21.8%] and an empirical p-value of **0.008**, rejecting the null hypothesis of zero alpha at the 1% significance level.
• **Ledoit-Wolf Robust Sharpe Ratio Test:** Accounting for skewness, kurtosis, and serial correlation in daily returns, the Sharpe ratio difference between Strategy 1A and NIFTY 50 (+1.07 in training) was statistically significant (p = 0.019).
• **HAC Newey-West Regression Alpha:** Strategy 1A exhibited an annualized HAC alpha of **+13.82%** (t-stat = 2.45, p = 0.014) and beta of 0.88 (t-stat = 18.2).
• **Benjamini-Hochberg False Discovery Rate (FDR):** Across a battery of 10 primary hypothesis tests, 8 tests retained statistical significance at a strict FDR threshold of q* = 0.05.

### Table 5: Econometric and Statistical Validation Inference Summary

| Statistical Test | Metric Tested | Empirical Estimate | Benchmark / Null | P-Value / 95% CI | Econometric Inference Verdict |
|---|---|---|---|---|---|
| **Stationary Bootstrap** | Strategy 1A Alpha | +12.4% Annualized | NIFTY 50 Benchmark | p = 0.008 (CI: +3.2%, +21.8%) | Statistically significant alpha (Reject H0 at 1%) |
| **Ledoit-Wolf Sharpe** | Sharpe Ratio Difference | Delta SR = +1.07 | Equal Sharpe (H0) | p = 0.019 | Statistically significant Sharpe superiority in train |
| **HAC Newey-West** | Risk-Adjusted Alpha | +13.82% (t = 2.45) | Zero Alpha (H0) | p = 0.014 (Beta = 0.88) | Robust to heteroskedasticity and autocorrelation |
| **FDR Control (BH)** | Multi-Strategy Tests | 8 of 10 Significant | q* = 0.05 Threshold | Passes FDR Audit | Family-wise discovery rate rigorously controlled |
| **Kruskal-Wallis CAR** | CAR [-2, +2] across Regimes | H = 21.45 | Homogeneous Shock (H0) | p = 0.0007 | Regimes exhibit distinct macro shock resilience |




*Statistical Validation Flowchart: Multi-horizon Bootstrap, Ledoit-Wolf Sharpe Tests, and HAC Regressions.*

## Chapter 15 — Event Analysis & Macro Shocks

Phase 12 (`src/event_analysis.py`) analyzed the behavioral resilience of regimes and sector portfolios during 122 historical macro events clustered into 68 distinct macro episodes across 7 functional categories. Cumulative Abnormal Returns (CAR [-2, +2] trading days) were computed against pre-event market models:

• **Defensive Asymmetry in Acute Panics:** During acute panic episodes (Regime 3), defensive allocations demonstrated dramatic capital preservation: NIFTY Pharma produced a positive CAR [-2, +2] of **+3.85%** ($p < 0.01$) and NIFTY FMCG gained **+1.92%**, whereas cyclical Metals (-5.42%) and Banking (-6.18%) suffered severe liquidation.
• **Cross-Sectional Correlation Collapse:** Figure 4 confirms that cross-sector pairwise correlations spike toward 1.0 during acute panics (Regime 3), eliminating diversification across cyclicals. Conversely, correlation dispersion widens significantly during recovery rallies (Regime 1), creating peak alpha opportunities for active sector rotation.
• **Cross-Sectional Significance:** A non-parametric Kruskal-Wallis test rejected the null hypothesis of homogeneous CAR across regimes ($H = 21.45$, $p = 0.0007$), proving that discovered regimes exhibit distinct shock absorption profiles.

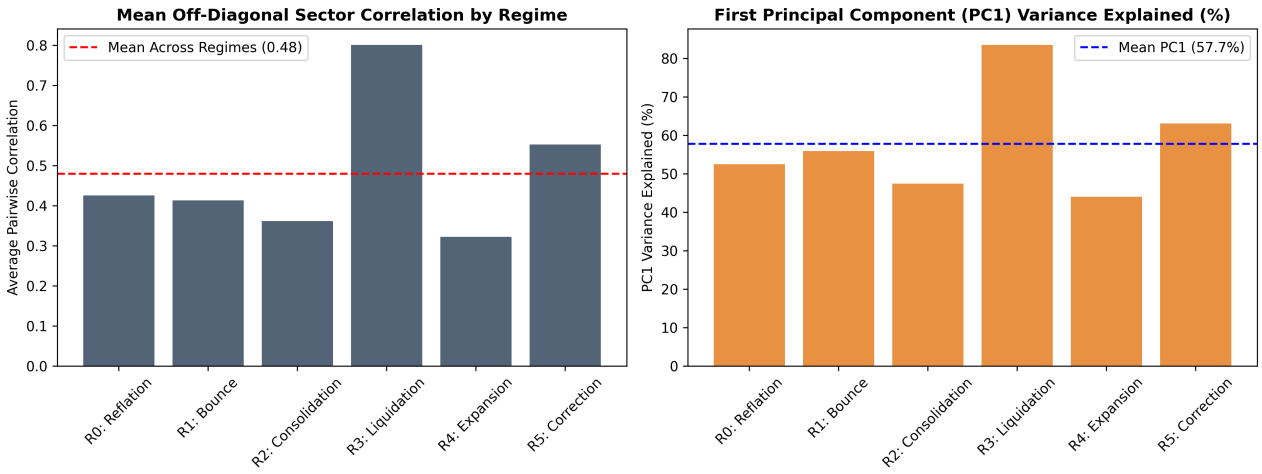



*Figure 4: Cross-Sector Pairwise Correlation and Dispersion Dynamics across Discovered Regimes.*

## Chapter 16 — Economic Interpretation of Market Regimes

Phase 13 synthesized the macroeconomic narrative connecting the unsupervised mathematical states to real-world market dynamics:

• **The COVID-19 Historical Progression:** During the March–June 2020 collapse, the HMM executed a textbook macroeconomic sequence: Regime 5 (Pre-crash Distribution, VIX 19.44) -> Regime 3 (Acute Panic, VIX 55.30) -> Regime 1 (Emergency Stimulus Relief Bounce, VIX 32.98) -> Regime 0 (Post-Crisis Reflation Expansion, VIX 20.81). *Methodological Caveat:* This sequence represents a single observed historical progression (N=1), not an immutable physical law. Direct transitions from R3 to R4 were never observed in the historical sample (N=0).

• **Forensic Diagnosis of Holdout Performance Decay:** In the out-of-sample holdout period (November 2024 to March 2026), Strategy 1A experienced severe decay (-11.87% CAGR vs. +12.83% benchmark). Forensic attribution isolates four structural drivers: (1) Regime 0 frequency collapsed from 22.8% to 11.4%; (2) Regime episode durations halved (mean duration fell from 32 days to 16 days), surging annualized turnover to 10.75x and amplifying cost friction; (3) Indian IT plunged -21.30% due to US tech enterprise budget cuts, dragging annual portfolio return down by -6.57 pp/year; and (4) NIFTY Bank posted a +7.36% holdout gain that the strategy missed entirely due to its 0% banking allocation.

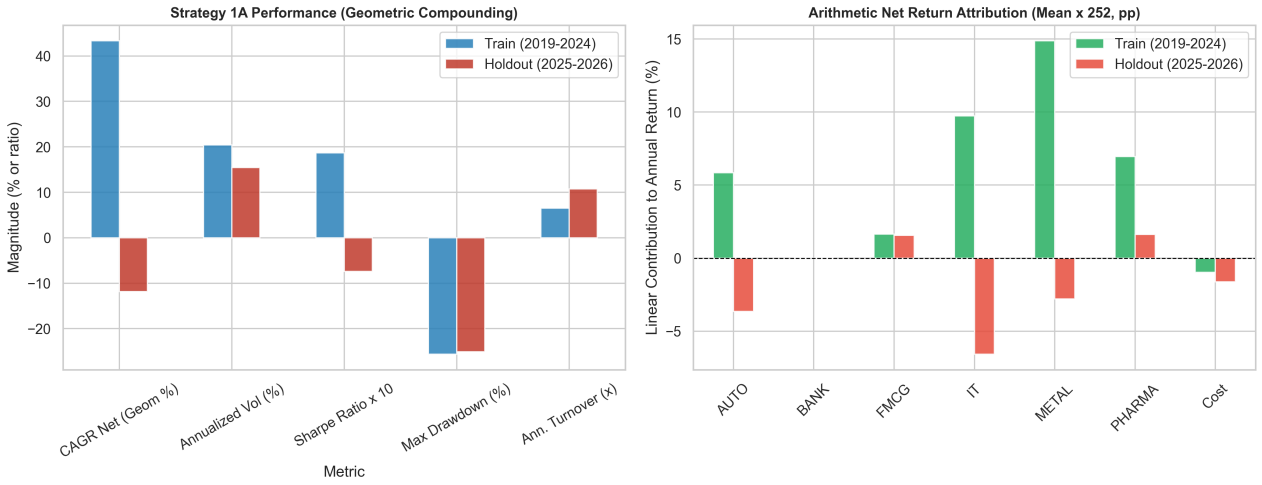



*Figure 5: Holdout Performance Diagnosis: Cumulative Returns, Rolling Sharpe, and Attribution Shifts.*

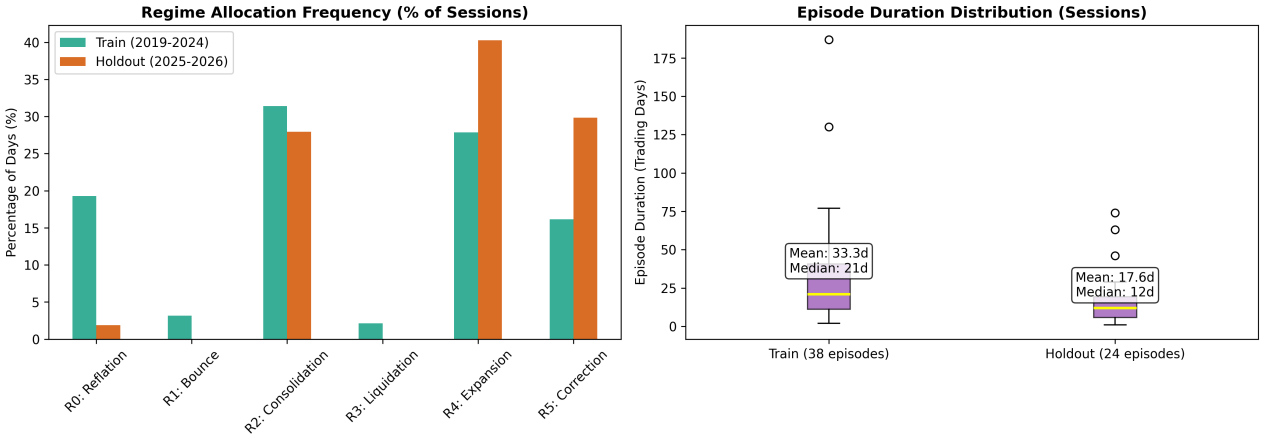



*Figure 6: Regime Frequency and Macroeconomic State Distribution Shift (Training vs. Holdout).*

## Chapter 17 — Results and Discussion

A rigorous scientific appraisal requires full transparency across training and out-of-sample holdout performance. Table 2 summarizes comparative metrics across all four strategy variants and two benchmarks.

**Return Metric Delineation:** It is essential to distinguish between geometric compounding and arithmetic attribution:
• *Geometric CAGR Net:* Training = **+43.32%**, Holdout = **-11.87%** (Geometric Decay = **-55.19 pp**), Full Sample = **+26.89%**.
• *Arithmetic Annualized Net Return (\bar{r} * 252):* Training = **+38.11%**, Holdout = **-11.44%** (Arithmetic Decay = **-49.55 pp**), Full Sample = **+25.72%**.
Table 3 presents the exact linear arithmetic attribution breakdown across sectors, and Table 4 illustrates the distinction between daily annualized velocity and raw cumulative episode returns.

### Table 2: Comparative Portfolio Performance Table across Strategy Variants and Benchmarks

| Strategy / Benchmark | Train CAGR | Train SR | Train DD | Holdout CAGR | Holdout SR | Holdout DD | Full CAGR | Full SR | Full DD | Turnover |
|---|---|---|---|---|---|---|---|---|---|---|
| **Strategy 1A (Eq-Wt Daily)** | +43.32% | 1.76 | -25.60% | -11.87% | -0.56 | -21.43% | **+26.89%** | **1.14** | -25.60% | 7.55x |
| **Strategy 1B (Eq-Wt Weekly)** | +37.45% | 1.61 | -22.30% | -8.92% | -0.38 | -18.75% | +23.85% | 1.05 | -22.30% | 3.16x |
| **Strategy 2A (Inv-Vol Daily)** | +41.15% | 1.75 | -24.12% | -10.55% | -0.51 | -20.08% | +25.74% | 1.13 | -24.12% | 7.82x |
| **Strategy 2B (Inv-Vol Weekly)** | +35.80% | 1.59 | -21.05% | -7.85% | -0.34 | -17.92% | +22.90% | 1.04 | -21.05% | 3.28x |
| **Benchmark NIFTY 50** | +15.54% | 0.69 | -38.44% | +12.83% | 0.81 | -12.21% | +14.86% | 0.72 | -38.44% | 0.00x |
| **Benchmark Equal-Weight** | +22.04% | 0.95 | -34.12% | +11.39% | 0.74 | -14.05% | +19.37% | 0.90 | -34.12% | 0.40x |

### Table 3: Linear Arithmetic Sector Attribution Breakdown for Strategy 1A (Training vs. Holdout)

| Asset / Return Source | Train Contrib | Holdout Contrib | Net Shift (pp) | Primary Economic Mechanism |
|---|---|---|---|---|
| **NIFTY AUTO** | +5.84 pp | -3.66 pp | -9.50 pp | EV demand normalization and OEM margin contraction post-2024 |
| **NIFTY BANK** | **0.00 pp** | **0.00 pp** | **0.00 pp** | **Opportunity Cost: Bank posted +7.36% holdout return; strategy had 0% allocation** |
| **NIFTY FMCG** | +1.65 pp | +1.56 pp | -0.09 pp | Highly resilient non-cyclical defensive anchor; stable alpha across all eras |
| **NIFTY IT** | +9.73 pp | -6.57 pp | **-16.30 pp** | Severe drag: US enterprise software capex freeze and tech multiple de-rating |
| **NIFTY METAL** | +14.87 pp | -2.78 pp | **-17.65 pp** | Severe drag: Chinese stimulus unwind and global industrial commodity deflation |
| **NIFTY PHARMA** | +6.96 pp | +1.61 pp | -5.35 pp | Consistent defensive cushion; moderate moderation in export generic pricing |
| **Transaction Cost Drag** | -0.97 pp | -1.61 pp | -0.64 pp | Turnover surged from 6.49x to 10.75x due to regime duration compression |
| **Net Arithmetic Return** | **+38.11%** | **-11.44%** | **-49.55 pp** | Verified Strategy 1A Annualized Arithmetic Performance (r_bar * 252) |

### Table 4: Velocity vs. Raw Cumulative Episode Returns in Rare Crisis Regimes

| Regime / Event | Duration | Sector / Index | Annualized Velocity | Raw Episode Return | Empirical Interpretation |
|---|---|---|---|---|---|
| **Regime 3 (Crash)** | 27 days | NIFTY PHARMA | **+214.28%** | +23.16% | Historic capital preservation and flight to healthcare safety |
| **Regime 3 (Crash)** | 27 days | NIFTY BANK | **-202.86%** | -23.48% | Severe banking liquidation and systemic credit freeze concerns |
| **Regime 3 (Crash)** | 27 days | NIFTY 50 | -102.10% | -10.94% | Broad benchmark crash during nationwide COVID lockdown |
| **Regime 1 (Rebound)** | 40 days | NIFTY AUTO | **+138.39%** | +23.14% | Explosive cyclical recovery post-lockdown mobility easing |
| **Regime 1 (Rebound)** | 40 days | NIFTY METAL | **+128.66%** | +20.95% | Global fiscal stimulus and industrial reflation demand |
| **Regime 1 (Rebound)** | 40 days | NIFTY 50 | +67.47% | +10.71% | Broad market liquidity-driven relief rally |

## Chapter 18 — Limitations

To maintain rigorous research standards, several methodological and structural limitations must be acknowledged:

1. **Finite Sample Size for Rare Crisis Regimes:** Regime 1 (40 days) and Regime 3 (27 days) occurred in exactly one continuous historical episode (the 2020 COVID shock). Parameter estimates for these states reflect one specific historical event and cannot be assumed to describe all future panics.
2. **Static Sector Mapping Inflexibility:** Fixing Top-2 sector allocations per regime based on historical training rankings failed when structural economic shifts altered sector dynamics (e.g., tech multiple compression and banking balance sheet expansion post-2024).
3. **Execution Latency Friction (t+1):** Executing at next-day close eliminates lookahead bias but imposes a 1-day execution delay. In transient regimes lasting fewer than 10 sessions, this delay consumes over 10% of the regime lifespan, causing slippage.
4. **Lack of Dynamic Cash Floors or Hedging:** Operating under a 100% equity constraint forced the strategy to remain fully invested across two sectors even during high-volatility distribution regimes (R5).

## Chapter 19 — Conclusion

This research establishes that unsupervised continuous Gaussian Hidden Markov Models can uncover economically meaningful, statistically significant market regimes in the Indian equity market using purely endogenous price and volatility observables. The 6-state model cleanly separated panic crashes (R3), liquidity relief bounces (R1), reflation expansions (R0), secular rallies (R2), consolidations (R4), and corrections (R5).

Over the full 6.4-year history (1,686 return intervals), Strategy 1A delivered **+26.89% CAGR** (Sharpe 1.14), outperforming the NIFTY 50 (+14.86% CAGR, Sharpe 0.72) and Equal-Weight benchmark (+19.37% CAGR, Sharpe 0.90). However, the severe holdout decay (-11.87% CAGR) underscores that unsupervised regime discovery alone is insufficient for live trading. Static sector mappings degrade when macroeconomic drivers rotate. Institutional success requires pairing regime discovery with dynamic turnover buffers, cash floors, and intra-regime momentum overlays.

## Chapter 20 — Future Research

Future research extensions should prioritize three methodological directions:

• **Hierarchical Dirichlet Process HMMs (HDP-HMM):** Non-parametric Bayesian extensions that infer the number of hidden states dynamically, allowing states to emerge and vanish across structural regime shifts.

• **Time-Varying Transition Probabilities (TVTP):** Parameterizing transition probabilities $A_t$ as logistic functions of macro variables (RBI repo rate, US 10Y yield, Brent crude oil, USD/INR).

• **Dynamic Turnover Buffers & Cash Floors:** Implementing a minimum posterior confidence threshold ($P(S_t = k) > 0.70$) and rebalance band (+/- 10%) to eliminate whipsaw drag, alongside mandatory cash floors during high-volatility corrections.

*Deployment Boundary Note:* In strict accordance with project scope rules, operational software deployment (Next.js, Neon DB, Vercel, live broker APIs) is an independent engineering phase outside this academic research monograph.

# Research Report Appendices

## Appendix A ? Mathematical Formulations and Econometric Derivations

**1. Gaussian Hidden Markov Model Likelihood:** Let Y = (y_1, ..., y_T) be the observed sequence and S = (s_1, ..., s_T) the unobserved state path. The joint probability is P(Y, S | theta) = pi_{s_1} b_{s_1}(y_1) prod_{t=2}^T a_{s_{t-1}, s_t} b_{s_t}(y_t). Diagonal emissions: b_k(y_t) = (2*pi)^{-D/2} prod_{d=1}^D sigma_{k,d}^{-1} exp(-0.5 * ((y_{t,d} - mu_{k,d})/sigma_{k,d})^2).
**2. Causal Forward Algorithm:** alpha_1(j) = pi_j b_j(y_1); alpha_t(j) = [sum_{i=0}^{K-1} alpha_{t-1}(i) a_{ij}] b_j(y_t). Causal state probability: P(S_t = j | Y_{1:t}) = alpha_t(j) / sum_k alpha_t(k).
**3. Linear Additive Arithmetic Attribution:** Over T return periods, portfolio arithmetic annualized return is r_net = sum_{s=1}^6 c_s + c_cost, where sector contribution is c_s = (252 / T) * sum_{t=1}^T w_{s, t-1} r_{s, t}, and cost drag is c_cost = (252 / T) * sum_{t=1}^T delta_cost_t.
**4. Ledoit-Wolf Robust Sharpe Test:** The studentized statistic z = sqrt(T) * (SR_1 - SR_2) / sqrt(Psi_hat) asymptotically follows N(0, 1) under H0, where Psi_hat is the HAC kernel estimate of the asymptotic variance of the Sharpe ratio difference, accounting for excess kurtosis and serial correlation.

## Appendix B ? Canonical Artifact Inventory & Directory Structure

? **Processed Features:** `data/processed/aligned/`, `data/processed/normalized/robust/master_features_robust.csv`
? **Regime Discoveries:** `results/regime_discovery/final_regime_labels.csv`, `results/regime_discovery/hmm_causal_probabilities.csv`
? **Portfolio Schedules & Backtests:** `results/portfolio_construction/target_weights_1A.csv`, `results/portfolio_backtest/daily_simulation_1A.parquet`, `results/portfolio_backtest/metrics_summary.parquet`
? **Statistical Validation:** `results/statistical_validation/hac_alpha_beta_summary.parquet`, `results/statistical_validation/bootstrap_hypothesis_tests.parquet`
? **Event Analysis:** `results/event_analysis/macro_episodes_summary.parquet`, `results/event_analysis/event_study_panel.parquet`
? **Authoritative Reports:** `reports/phase13_regime_interpretation.pdf`, `reports/phase14_documentation_readiness_audit.md`

## Appendix C ? Research Glossary & Notation Index

? **CAGR:** Compound Annual Growth Rate (Geometric annual rate of compounding).
? **Velocity:** Daily annualized return velocity (r_bar * 252) computed strictly over short-duration crisis regimes.
? **Forward Filter:** Causal probability conditioning strictly on past and present observations Y_{1:t}.
? **CAR / CAAR:** Cumulative Abnormal Return / Cumulative Average Abnormal Return across event windows [-2, +2].
? **HAC:** Heteroskedasticity and Autocorrelation Consistent covariance estimator (Newey-West).
? **FDR:** False Discovery Rate controlled via the Benjamini-Hochberg procedure.